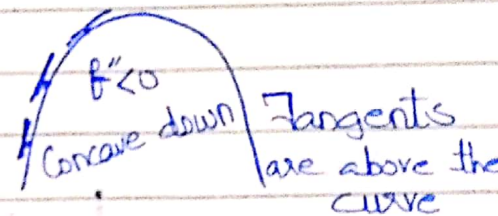
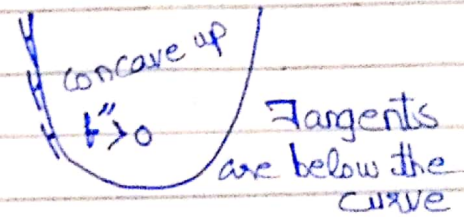
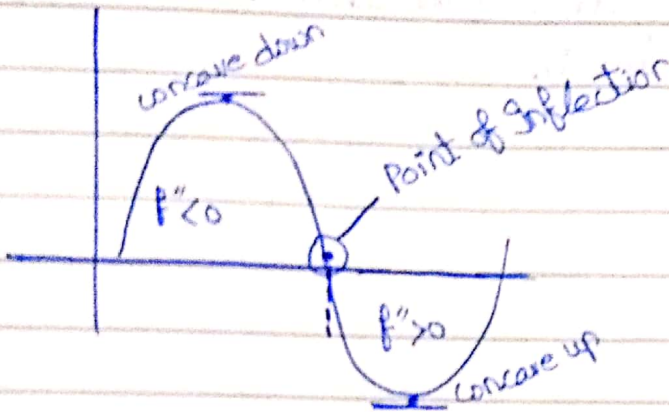


Sec 4.4 Concavity & Curve Sketching:



Point of Inflection:

A point where concavity changes is called point of Inflection

At a point of Inflection, either $f'' = 0$ or fail to exist

Theorem (2nd derivative test for concavity)

Let $y = f(x)$ be twice differentiable function on an Interval I

- (a) If $f'' > 0$ on I , then $f(x)$ is concave up on I .
- (b) If $f'' < 0$ on I , then $f(x)$ is concave down on I .

Example:

$$f(x) = x^2$$

$$D_f = (-\infty, \infty)$$

Find the ~~function~~ ^{interval} where the function is concave up or concave down.

To identify point of Inflection

$$f' = 2x$$

$$f'' = 2$$

f'' cannot be zero or undefined for any value of x . There is no point of inflection.

Interval $(-\infty, \infty)$

- | | | |
|-----------------------|--|--|
| • Sign of f'' | | $f'' > 0$ (for any x)
Concave up |
| • Behaviour of $f(x)$ | | |

Example:

$$f(x) = x^3$$

$$D_f = (-\infty, \infty)$$

Find interval where function is concave up or concave down.

To identify point of inflection

$$f' = 3x^2$$

$$f'' = 6x$$

$x=0$ is a point of Inflection

$(-\infty, 0)$

$(0, \infty)$

- | | | |
|-----------------------|--|-------------------------------|
| • Sign of f'' | | $f''(-1) < 0$
Concave down |
| • Behaviour of $f(x)$ | | |

- | | | |
|-----------------------|--|----------------------------|
| • Sign of f'' | | $f''(1) > 0$
Concave up |
| • Behaviour of $f(x)$ | | |

To find critical point

$$f'(x) = 3x^2$$

at $x=0$

$(-\infty, 0)$

$(0, \infty)$

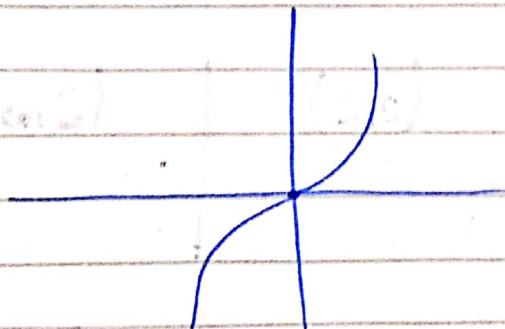
• Sign of f'
• Behaviour of $f(x)$

$f' > 0$
 f is increasing

• Sign of f'
• Behaviour of $f(x)$

$f' > 0$
 f is increasing

at $x=0$, f' does not change its sign, so neither local maxima nor local minima.



*Example:

$$\frac{x^4}{4} - 2x^2 + 4$$

(Imp)

- Identify where the extreme values of $f(x)$ occurs
- Find the intervals on which f is increasing and f is decreasing
- Find where graph of $f(x)$ is concave up and where it is concave down.
- Sketch the general shape of graph for $f(x)$
(Plot this graph by using local maxima and local minima, point of inflection and intercepts)

Q. 17. Find
the extrema of the function

$$f(x) = \frac{x^4}{4} - 2x^2 + 4$$

$$D_f(-\infty, \infty)$$

Sol:

(a) $f'(x) = x^3 - 4x$

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$x = 0, \quad x = \pm 2$$

So, $x = -2, 0, 2$ are three critical points

i.e.,

extremas appear at $x = -2, 0, 2$

(b)

$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
• $f'(-3) < 0$	• $f'(-1) > 0$	• $f'(1) < 0$	• $f' > 0$
• Decreasing	• Increasing	• Decreasing	• Increasing

• at $x = -2$

$$\begin{array}{ccccccc} -\infty & & -2 & & 0 & & 2 & & \infty \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & f' < 0 & & f' > 0 & & f' < 0 & & f' > 0 \end{array}$$

$f'(x)$ changes from -ve to +ve. So, there is local minima

$$f(-2) = 0$$

• at $x = 0$

" " local maxima

$$f(0) = 4$$

• at $x = 2$

" " local minima

$$f(2) = 0$$

(c) $f'' = 3x^2 - 4$

$$3x^2 - 4 = 0$$

$$x = \pm \frac{2}{\sqrt{3}}$$

$$-\infty \mid -\frac{2}{\sqrt{3}} \quad \frac{2}{\sqrt{3}} \mid \infty$$

$$(-\infty, -\frac{2}{\sqrt{3}})$$

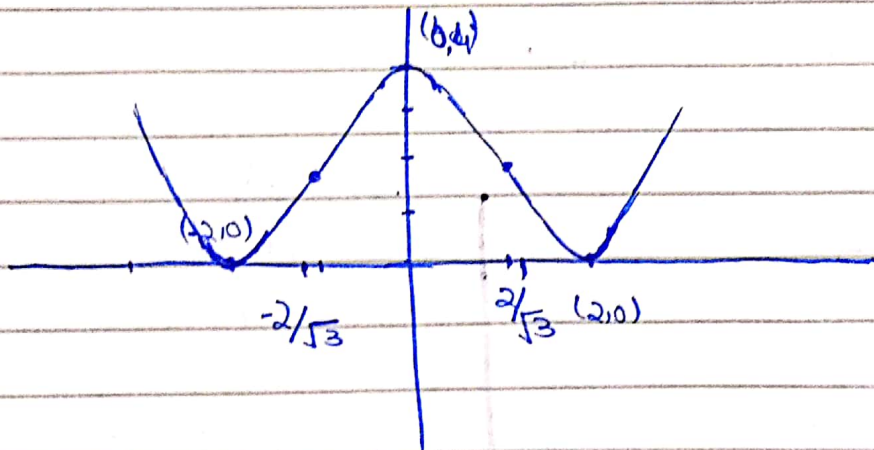
$$(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$$

$$(\frac{2}{\sqrt{3}}, \infty)$$

$f''(-2) > 0$
Concave up

$f''(0) < 0$
Concave down

$f''(2) > 0$
Concave up



Q $f(x) = \sin|x|$; $[-2\pi, 2\pi]$

- Identify the local extremas
- Find the interval on which f is increasing/decreasing
- Find where $f(x)$ is concave up / concave down
- Sketch the graph of $f(x)$
(by identify local maxima, inflection points & intercepts)

Sol:

$$|x| = \begin{cases} -x & ; x < 0 \\ x & ; x \geq 0 \end{cases}$$

$$\sin|x| = \begin{cases} \sin(-x) & ; -2\pi \leq x < 0 \\ \sin(x) & ; 0 \leq x \leq 2\pi \end{cases}$$

Case - I :

where $-2\pi \leq x < 0$

$$f(x) = -\sin x$$

$$f'(x) = -\cos x$$

$$-\cos x = 0$$

$$x = -3\pi/2, -\pi/2$$

So, critical points are at

$$x = -3\pi/2, -\pi/2$$

Case - II

where $0 \leq x \leq 2\pi$

$$f(x) = \sin(x)$$

$$f'(x) = \cos x$$

$$\cos x = 0$$

$$x = 3\pi/2, \pi/2$$

So, critical points are at

$$x = 3\pi/2, \pi/2$$

Intervals	$(-2\pi, -3\pi/2)$	$(-3\pi/2, -\pi/2)$	$(-\pi/2, 0)$	Intervals	$(0, \pi/2)$	$(\pi/2, 3\pi/2)$	$(3\pi/2, 2\pi)$
Sign of $f'(x)$	$f' < 0$	$f' > 0$	$f' < 0$	Signs of $f'(x)$	$f' > 0$	$f' < 0$	$f' > 0$
Behaviour	decreasing	increasing	decreasing	Behaviour	Increasing	Decreasing	increasing

at $x = -3\pi/2$, f' changes from -ve to +ve, So there is local minima

$$f(-3\pi/2) = -1 \quad (x, y) = (-3\pi/2, -1)$$

at $x = -\pi/2$, " local maxima

$$f(-\pi/2) = 1 \quad (x, y) = (-\pi/2, 1)$$

at $x = \pi/2$, " local maxima

$$f(\pi/2) = 1 \quad (x, y) = (\pi/2, 1)$$

at $x = 3\pi/2$, " local minima

$$f(3\pi/2) = -1 \quad (x, y) = (3\pi/2, -1)$$

(c)

$$-2\pi \leq x < 0$$

$$f(x) = -\sin x$$

$$f'(x) = -\cos x$$

$$f''(x) = \sin x$$

$$\sin x = 0$$

$$x = -\pi, -2\pi$$

$$0 \leq x \leq 2\pi$$

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$-\sin x = 0$$

$$x = 0, \pi, 2\pi$$

Intervals	$(-2\pi, -\pi)$	$(-\pi, 0)$	Interval	$(0, \pi)$	$(\pi, 2\pi)$
Sign of f''	$f'' > 0$	$f'' < 0$	Sign of f''	$f'' > 0$ $f'' < 0$	$f'' > 0$
Behaviour of $f(x)$	Concave up	Concave down	Behaviour of $f(x)$	Concave down	Concave up

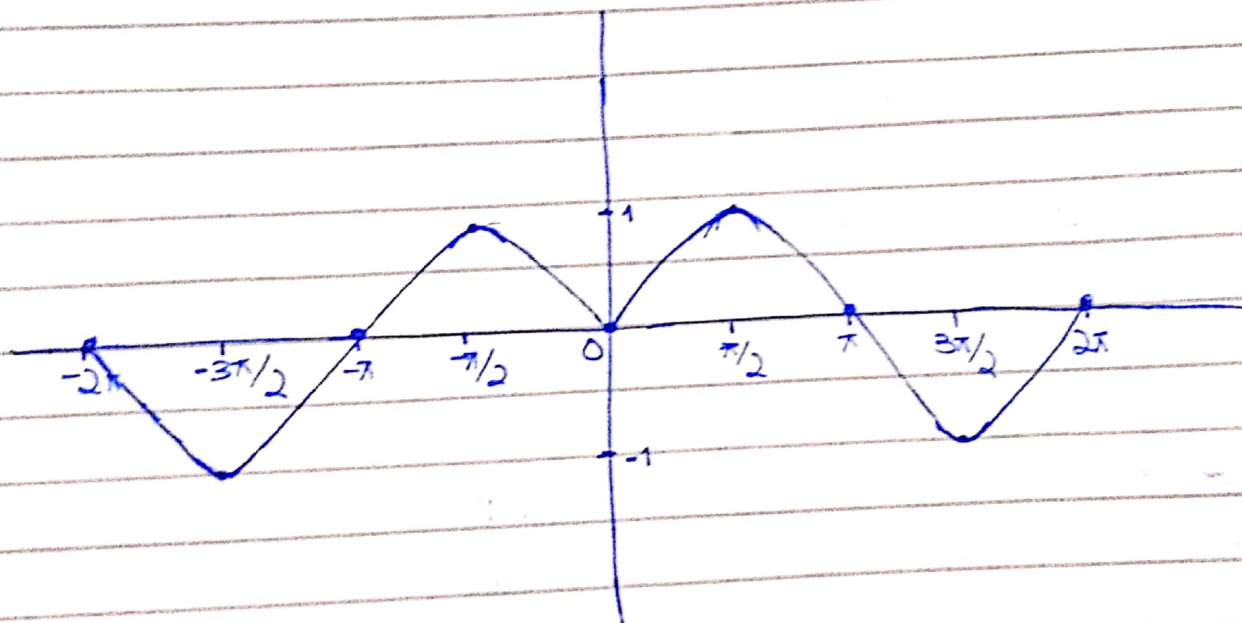
$$-2\pi \Rightarrow f(-2\pi) = 0 \Rightarrow (x, y) = (-2\pi, 0)$$

$$-\pi \Rightarrow f(-\pi) = 0 \Rightarrow (x, y) = (-\pi, 0)$$

$$0 \Rightarrow f(0) = 0 \Rightarrow (x, y) = (0, 0)$$

$$\pi \Rightarrow f(\pi) = 0 \Rightarrow (x, y) = (\pi, 0)$$

$$2\pi \Rightarrow f(2\pi) = 0 \Rightarrow (x, y) = (2\pi, 0)$$



$$f(x) = \sin x$$

$$(-3\pi/2, -1), (-\pi/2, 1) \text{ C.pts}$$

$$(-2\pi, 0), (-\pi, 0) \text{ Inf. pts}$$

$$f(x) = \sin x$$

$$(\pi/2, 1), (3\pi/2, -1) \text{ C.pts}$$

$$(0, 0), (\pi, 0), (2\pi, 0) \text{ Inf. pts}$$

Point $x=c$ such that $f'(c)=0$
is called **stationary point**

Theorem: (2nd derivative test for local extrema)

Suppose that $y=f(x)$ is twice differentiable on an interval I , containing $x=c$, such that $f'(c)=0$

- (a) If $f''(c) > 0$ on I , then ~~$f(x)$ is concave down~~ at $x=c$, $f(x)$ has local minima
- (b) If $f''(c) < 0$ on I , then at $x=c$, $f(x)$ has local maxima
- (c) If $f''(c) = 0$, then test fails.

Integration:

Integrals

Indefinite Integrals
(Integrals without limits)

Definite Integrals
(Integrals with limits)

Proper Integrals
e.g.
(1) $\int_a^b f(x) dx$
lower limit $\leftarrow a$ $\rightarrow$ upper limit b
(2) $\int_0^\pi \sin x dx$

Improper Integrals
e.g.
(1) $\int_a^\infty f(x) dx$
(2) $\int_b^\infty f(x) dx$
(3) $\int_{-\infty}^\infty f(x) dx$

Integration by parts:

let u & v be two functioning of x

i.e) $u = u(x)$
 $v = v(x)$

$$\int u \cdot v dx = u \cdot \int v dx - \int \left(\int v dx \cdot u' \right) dx$$

Examples:

$$\textcircled{1} \int \sin x \, dx = -\cos x + C$$

$$\begin{aligned} \textcircled{2} \int \sin^2 x \, dx &= \int \frac{1 - \cos 2x}{2} \, dx \\ &= \frac{1}{2} \int (1 - \cos 2x) \, dx \\ &= \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) + C \end{aligned}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

OR

$$\int \sin^2 x \, dx = \int \sin x \cdot \sin x \, dx$$

$$\cos^2 x = 1 - \sin^2 x$$

$$= \sin x \cdot \int \sin x \, dx - \int \left(\int \sin x \, dx \cdot \frac{d}{dx} (\sin x) \right) dx$$

$$= -\sin x \cdot \cos x + \int \cos^2 x \, dx$$

$$= -\sin x \cdot \cos x + \int (1 - \sin^2 x) \, dx$$

$$\overset{I}{\int} \sin^2 x \, dx = -\sin x \cdot \cos x + \int 1 \, dx - \int \sin^2 x \, dx$$

$$2I = -\sin x \cdot \cos x + x + C$$

$$I = \frac{1}{2} (-\sin x \cdot \cos x + x + C)$$

$$\textcircled{3} \int \sin^3 x \, dx$$

$$= \int \overset{(n-1)}{\sin^2 x} \cdot \overset{(n)}{\sin x} \, dx$$

$$= -\sin^2 x \cos x - \int -\cos x \cdot 2\sin x \cos x \, dx$$

$$= -\sin^2 x \cos x + 2 \int \sin x \cdot \cos^2 x \, dx$$

$$= -\sin^2 x \cos x + 2 \int \sin x (1 - \sin^2 x) \, dx$$

$$I = -\sin^2 x \cos x + 2 \int \sin x \, dx - 2 \int \sin^3 x \, dx$$

$$3I = -\sin^2 x \cos x + 2 \int \sin x \, dx$$

$$3I = -\sin^2 x \cos x - 2 \cos x + C$$

$$I = -\frac{1}{3} \sin^2 x \cos x - \frac{2}{3} \cos x + C$$

$$\textcircled{4} \int \sin^4 x \, dx$$

$$= \int \overset{u}{\sin^3 x} \cdot \overset{v}{\sin x} \, dx$$

$$= -\sin^3 x \cos x - \int -\cos x \cdot 3\sin^2 x \cos x \, dx$$

$$= -\sin^3 x \cos x + 3 \int \sin^2 x \cdot \cos^2 x \, dx$$

$$= -\sin^3 x \cos x + 3 \int \sin^2 x (1 - \sin^2 x) \, dx$$

$$I = -\sin^3 x \cos x + 3 \int \sin^2 x \, dx - 3 \int \sin^4 x \, dx$$

$$4I = -\sin^3 x \cos x + 3 \int \sin^2 x \, dx$$

$$4I = -\sin^3 x \cos x + 3 \left(\frac{1 - \cos 2x}{2} \right)$$

$$I = \frac{1}{4} \left(-\sin^3 x \cos x + \frac{3}{2} (1 - \cos 2x) \right)$$

$$\int \sec x \, dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$$

$$= \ln |\sec x + \tan x| + c$$

$$\int f'(x) \cdot [f(x)]^n \, dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

$$\int \frac{1}{x} \cdot [\ln x]^2 \, dx = \frac{[\ln x]^3}{3} + c$$

$$\int f'(x) \cdot e^{f(x)} \, dx = e^{f(x)} + c$$

$$\frac{1}{2} \int 2x e^{x^2} \, dx = \frac{1}{2} e^{x^2} + c$$

$$\int \sec^3 x \, dx = \int \underbrace{\sec x}_u \cdot \underbrace{\sec^2 x}_v \, dx$$

$$\tan x \quad \frac{d \tan x}{dx} = \sec^2 x$$

$$= \sec x \cdot \tan x - \int \tan^2 x \cdot \sec x \, dx$$

$$= \sec x \cdot \tan x - \int \sec x (\sec^2 x - 1) \, dx$$

$$I = \sec x \cdot \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$2I = \sec x \cdot \tan x + \int \sec x \, dx$$

$$I = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + c$$

$$\int \csc^3 x \, dx$$

$$\int \csc^4 x \, dx$$

Ex

① $\int x^2 \cos x \, dx$

② $\int x^2 e^{2x} \, dx$

③ $\int e^{2x} \cos 2x \, dx$

④ $\int \ln x$

⑤ $\int x^2 \ln x \, dx$

① $x^2 \int \cos x \, dx - \int \int \cos x \, dx \frac{d}{dx} x^2 \, dx$

$$x^2(\sin x) - \int (\sin x)(2x) \, dx$$

$$x^2(\sin x) - 2 \int x \sin x \, dx$$

$$x^2(\sin x) - 2 \left[(x) \int \sin x \, dx - \int \int \sin x \frac{d}{dx} x \, dx \right]$$

$$x^2(\sin x) - 2 \left[x \cos x - \cos x \right] + c$$

③ $\cos 2x \int e^{2x} \, dx - \int \int e^{2x} \, dx$

$$= \cos 2x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} (-2 \sin 2x) \, dx$$

$$= \cos 2x \frac{e^{2x}}{2} + \int e^{2x} \sin 2x \, dx$$

$$= \cos 2x \frac{e^{2x}}{2} + \left(\sin 2x \int e^{2x} \, dx - \int \int e^{2x} \frac{d}{dx} \sin 2x \, dx \right)$$

$$= \cos 2x \frac{e^{2x}}{2} + \left(\sin 2x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} 2 \cos 2x \, dx \right)$$

$$= \cos 2x \frac{e^{2x}}{2} + \left[\sin 2x \frac{e^{2x}}{2} - \int e^{2x} \cos 2x \, dx \right]$$

$$I = \cos 2x \frac{e^{2x}}{2} + \sin 2x \frac{e^{2x}}{2} - I + C$$

$$2I = \cos 2x \frac{e^{2x}}{2} + \sin 2x \frac{e^{2x}}{2} + C$$

$$I = \frac{1}{2} \cos 2x \frac{e^{2x}}{2} + \frac{1}{2} \sin 2x \frac{e^{2x}}{2} + C$$

$$\begin{aligned} \textcircled{4} \int 1 \cdot \ln x \, dx &= \ln x \cdot \int (1) \, dx - \int (1) \, dx \cdot \frac{d}{dx} (\ln x) \\ &= \ln x \cdot x - \int x \cdot \frac{1}{x} \, dx \\ &= x \ln x - \int 1 \, dx \\ &= x \ln x - x + C \end{aligned}$$

$$\begin{aligned} \int \sin(\ln x) \, dx &= \sin(\ln x) x - \int x \cos(\ln x) \cdot \frac{1}{x} \, dx \\ &= x \sin(\ln x) - \int 1 \cdot \cos(\ln x) x \, dx \\ &= x \sin(\ln x) - \int \cos(\ln x) \cdot x - \int x \sin(\ln x) \cdot \frac{1}{x} \, dx \\ &= x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) \, dx + C \\ 2I &= x \sin(\ln x) - x \cos(\ln x) + C \end{aligned}$$

Guidelines for Evaluating

$$\int \sin^m x \cdot \cos^n x \, dx \quad \text{where } m \text{ \& } n \text{ are +ve integers}$$

Case-I

If m is an odd integer, then

$$\int \sin^m x \cdot \cos^n x \, dx = \int \sin^{m-1} x \cdot \sin x \cdot \cos^n x \, dx$$

- * Convert $\sin^{m-1} x$ into $\sin^2 x$ of $\cos x$ by using $\sin^2 x = 1 - \cos^2 x$
- * Use substitution

$$u = \cos x$$

$$du = -\sin x \, dx$$

e.g: ① $\int \sin^5 x \cdot \cos^6 x \, dx$

$$= \int \sin^4 x \cdot \sin x \cdot \cos^6 x \, dx$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\sin^4 x = (1 - \cos^2 x)^2$$

$$= \int (1 - \cos^2 x)^2 \cdot \sin x \cdot \cos^6 x \, dx$$

$$\text{let } u = \cos x$$

$$du = -\sin x \, dx \Rightarrow -du = \sin x \, dx$$

$$= - \int (1 - u^2)^2 u^6 \, du \quad (\because du = -\sin x \, dx)$$

$$= - \int (1 + u^4 - 2u^2) u^6 \, du$$

$$= - \int (u^6 + u^{10} - 2u^8) \, du$$

$$= - \left[\frac{u^7}{7} + \frac{u^{11}}{11} - \frac{2u^9}{9} \right] + C$$

$$\therefore u = \cos x$$

$$I = - \left[\frac{\cos^7 x}{7} + \frac{\cos^{11} x}{11} - \frac{2 \cos^9 x}{9} \right] + C$$

Evaluate

$$\textcircled{2} \int \sin^7 x \cdot \cos^3 x \, dx$$

$$= \int \sin^6 x \cdot \sin x \cdot \cos^3 x \, dx$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\sin^6 x = (1 - \cos^2 x)^3$$

$$= \int (1 - \cos^2 x)^3 \cdot \sin x \cdot \cos^3 x \, dx$$

$$\text{let } u = \cos x$$

$$du = -\sin x \, dx \Rightarrow -dx = \sin x \, dx$$

$$= - \int (1 - u^2)^3 u^3 \, du$$

$$= - \int$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$n \neq -1$

Case-II

If n is an odd integer

$$\int \sin^n x \cdot \cos^n x \, dx = \int \sin^n x \cdot \cos^{n-1} x \cdot \cos x \, dx$$

* Convert $\cos^{n-1} x$ into terms of $\sin x$ by using $\cos^2 x = 1 - \sin^2 x$

* let $u = \sin x$
 $du = \cos x \, dx$

Evaluate:

$$\textcircled{3} \int \sin^7 x \cdot \cos^3 x \, dx$$

$$= \int \sin^6 x (1 - \sin^2 x) \cdot \cos x \, dx$$

$$\text{let } u = \sin x \\ du = \cos x \, dx$$

$$= \int u^6 (1 - u^2) \, du$$

$$= \int (u^6 - u^8) \, du = \left(\frac{u^7}{7} - \frac{u^9}{9} \right) + C = \left(\frac{(\sin x)^7}{7} - \frac{(\sin x)^9}{9} \right) + C$$

Case-III

If m & n both are even ~~there~~ there is no standard method for it you can choose any method of your own choice

$$\text{e.g. } \int \sin^2 x \cdot \cos^2 x \, dx$$

$$= \int \left(\frac{1 - \cos 2x}{2} \right) \cdot \left(\frac{1 + \cos 2x}{2} \right) \, dx$$

$$= \frac{1}{2} \int (1 - \cos^2 2x) \, dx$$

$$= \frac{1}{2} \int 1 - \left(\frac{1}{2} + \frac{\cos 4x}{2} \right) \, dx$$

$$= \frac{1}{2} \int \frac{1}{2} - \frac{\cos 4x}{2} \, dx$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\cos^2 2x = \frac{1 + \cos 4x}{2}$$

$$= \frac{1}{4} \int (1 - \cos 4x) dx$$

$$= \frac{1}{4} \left[x - \frac{\sin 4x}{4} \right] + C$$

$$\int \tan^m x \sec^n x dx$$

Case-I

① If m is an odd; Write the integral as

$$\int \tan^m x \sec^n x dx = \int \tan^{m-1} x \cdot \tan x \cdot \sec^n x dx$$

and express $\tan^{m-1} x$ in terms of $\sec x$ by using trigonometric identity $\tan^2 x = \sec^2 x - 1$. Make this substitution

$$\text{let } u = \sec x$$

$$du = \sec x \tan x dx$$

$$\text{e.g.} = \int \tan^5 x \sec^4 x dx$$

$$= \int \tan^4 x \cdot \tan x \sec^4 x dx$$

$$= \int (\sec^2 x - 1)^2 \cdot \tan x \cdot \sec x \cdot \sec^3 x dx$$

$$\because \tan^2 x = \sec^2 x - 1$$

$$\because u = \sec x$$

$$du = \sec x \tan x dx$$

$$= \int (u^2 - 1)^2 u^3 du$$

$$\begin{aligned}
 &= \int (u^4 - 2u^2 + 1) u^3 dx \\
 &= \int (u^7 - 2u^5 + u^3) dx \\
 &= \int u^7 du - 2 \int u^5 du + \int u^3 du
 \end{aligned}$$

Case-II

gf n is an even integer

$$\int \tan^m x \sec^n x dx = \int \tan^m x \sec^{n-2} x \sec^2 x dx$$

Express $\sec^2 x = 1 + \tan^2 x$

let $u = \tan x$

$du = \sec^2 x dx$

e.g. $\int \tan^5 x \sec^4 x dx$

$$\begin{aligned}
 &= \int \tan^5 x \cdot \sec^2 x \cdot \sec^2 x dx \\
 &= \int \tan^5 x \cdot (1 + \tan^2 x) \cdot \sec^2 x dx \\
 \text{let } &u = \tan x \\
 &du = \sec^2 x dx \\
 &= \int u^5 \cdot (1 + u^2) du \\
 &= \int (u^5 + u^7) du
 \end{aligned}$$

Case-III

gf m is even and n is odd. There is no standard method of evaluating. Possible use integration by parts.

Trigonometric Substitutions

Expression in Integrand

Substitution

① $\sqrt{a^2 - x^2}$

$x = a \sin \theta$

② $\sqrt{a^2 + x^2}$

$x = a \tan \theta$

③ $\sqrt{x^2 - a^2}$

$x = a \sec \theta$

Ex Evaluate:

$$\int \frac{1}{x^2 \sqrt{16 - x^2}} dx$$

$$\int \frac{1}{x^2 \sqrt{4^2 - x^2}} dx$$

let $x = a \sin \theta$

$a = 4$

$x = 4 \sin \theta$

$dx = 4 \cos \theta d\theta$

$$\int \frac{1}{4^2 \sin^2 \theta \sqrt{4^2 - 4^2 \sin^2 \theta}} \cdot 4 \cos \theta d\theta$$

$$\int \frac{1}{4^2 \sin^2 \theta \cdot 4 \sqrt{1 - \sin^2 \theta}} \cdot 4 \cos \theta d\theta$$

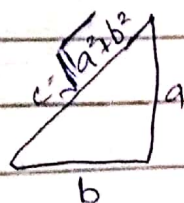
$$\int \frac{1}{4^2 \sin^2 \theta \cdot 4 \sqrt{\cos^2 \theta}} d\theta$$

$$= \frac{1}{16} \int \operatorname{cosec}^2 \theta d\theta$$

$$I = -\frac{1}{16} \cot \theta + c = \boxed{-\frac{1}{16} \frac{\sqrt{4^2 - x^2}}{x} + c}$$

$x = 4 \sin \theta$

$\sin \theta = \frac{x}{4}$



$b = \sqrt{c^2 - a^2}$

$b = \sqrt{4^2 - x^2}$

So, $\cos \theta = \frac{\sqrt{4^2 - x^2}}{4}$

$\cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\sqrt{4^2 - x^2}}{x}$

Evaluate

$$\int \frac{1}{\sqrt{4x^2 - 25}} dx$$

$$\int \frac{1}{\sqrt{(2x)^2 - (5)^2}} dx = \int \frac{1}{2\sqrt{x^2 - (5/2)^2}} = \frac{1}{2} \int \frac{1}{x^2 - (5/2)^2}$$

let $x = a \sec \theta$

$$a = 5/2$$

$$x = \frac{5}{2} \sec \theta$$

$$dx = \frac{5}{2} \sec \theta \cdot \tan \theta d\theta$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{(5/2)^2 \sec^2 \theta - (5/2)^2}} \cdot \frac{5}{2} \sec \theta \tan \theta d\theta$$

$$= \frac{1}{2} \cdot \frac{5/2}{5/2} \int \frac{1}{\sqrt{\sec^2 \theta - 1}} \cdot \sec \theta \tan \theta d\theta$$

$$= \frac{1}{2} \int \frac{1}{\tan \theta} \cdot \sec \theta \tan \theta d\theta$$

$$= \frac{1}{2} \int \sec \theta d\theta$$

$$= \frac{1}{2} \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{1}{2} \ln \left| \frac{x}{5/2} + \frac{\sqrt{x^2 - (5/2)^2}}{5/2} \right| + C$$

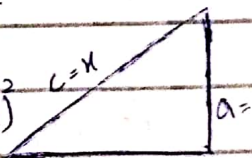
$$x = 5/2 \sec \theta$$

$$\sec \theta = \frac{x}{5/2}$$

$$\cos \theta = \frac{5/2}{x}$$

$$a = \sqrt{c^2 - b^2}$$

$$= \sqrt{x^2 - (5/2)^2}$$



$$\sin \theta = \frac{\sqrt{x^2 - (5/2)^2}}{x}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{x^2 - (5/2)^2}}{5/2}$$

$$I = \int \frac{(4+k^2)^2}{k^3} dk$$

$$\text{or } I = \int \frac{(2^2+k^2)^2}{k^3} dk$$

$$\begin{aligned} \text{let } k &= a \tan \theta \\ k &= 2 \tan \theta \quad a=2 \\ dk &= 2 \sec^2 \theta \cdot d\theta \end{aligned}$$

$$I = \int \frac{(2^2 + (2 \tan \theta)^2)^2}{(2 \tan \theta)^3} \cdot 2 \sec^2 \theta \cdot d\theta$$

$$= \int \frac{(4 + 4 \tan^2 \theta)^2}{8 \tan^3 \theta} \cdot 2 \sec^2 \theta \cdot d\theta$$

$$= \int \frac{\cancel{16} (1 + \tan^2 \theta)^2}{\cancel{8} \tan^3 \theta} \cdot 2 \sec^2 \theta \cdot d\theta$$

$$= \int \frac{4 (\sec^2 \theta)^2}{\tan^3 \theta} \cdot \sec^2 \theta \cdot d\theta$$

$$= 4 \int \frac{\sec^6 \theta}{\tan^3 \theta} d\theta$$

$$= 4 \int \frac{\frac{1}{\cos^6 \theta} \cdot \frac{1}{\cos^3 \theta}}{\frac{\sin^3 \theta}{\cos^3 \theta}} d\theta$$

$$= 4 \int \frac{1}{\sin^3 \theta \cos^3 \theta} d\theta$$

$$= 8 \times 4 \int \frac{1}{(2 \sin \theta \cos \theta)^3} d\theta$$

$$= 32 \int \frac{1}{(\sin 2\theta)^3} d\theta$$

$$= 32 \int \operatorname{cosec}^3 2\theta d\theta$$

$$I = 32 \int \underbrace{\operatorname{cosec}^2 2\theta}_u \cdot \underbrace{\operatorname{cosec} 2\theta}_v d\theta$$

∴ Integration by parts

$$= 32 \left(-\frac{1}{2} \operatorname{cosec} 2\theta \cdot \cot 2\theta - \int -\frac{1}{2} \cot 2\theta \cdot -2 \operatorname{cosec} 2\theta \cdot \cot 2\theta d\theta \right)$$

$$= 32 \left(-\frac{1}{2} \operatorname{cosec} 2\theta \cdot \cot 2\theta - \int \cot^2 2\theta \cdot \operatorname{cosec}^2 2\theta \cdot d\theta \right)$$

$$\therefore \cot^2 2\theta = \operatorname{cosec}^2 2\theta - 1$$

$$= 32 \left(-\frac{1}{2} \operatorname{cosec} 2\theta \cdot \cot 2\theta - \int (\operatorname{cosec}^2 2\theta - 1) \operatorname{cosec} 2\theta d\theta \right)$$

$$= 32 \left(-\frac{1}{2} \operatorname{cosec} 2\theta \cdot \cot 2\theta - \int \operatorname{cosec}^3 2\theta d\theta + \int \operatorname{cosec} 2\theta d\theta \right)$$

$$2I = 32 \left(-\frac{1}{2} \operatorname{cosec} 2\theta \cdot \cot 2\theta + \ln |\operatorname{cosec} 2\theta - \cot 2\theta| + c \right)$$

$$I = 32 \left(-\frac{1}{4} \operatorname{cosec} 2\theta \cdot \cot 2\theta + \ln |\operatorname{cosec} 2\theta - \cot 2\theta| + c \right)$$

Evaluate:

$$\int \frac{4x^2 + 13x - 9}{x^2 + 2x^2 - 3x} dx$$

$$\frac{4x^2 + 13x - 9}{x^2 + 2x^2 - 3x} = \frac{4x^2 + 13x - 9}{x(x+3)(x-1)}$$

$$\frac{4x^2 + 13x - 9}{x(x+3)(x-1)} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-1}$$

$$4x^2 + 13x - 9 = A(x+3)(x-1) + B(x)(x-1) + C(x)(x+3)$$

Evaluate:

$$\textcircled{1} \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

$$\textcircled{2} \int f'(x) [f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

Evaluate:

$$I = \int \frac{x^2 - x - 21}{2x^3 - x^2 + 8x - 4} dx$$

$$\frac{x^2 - x - 21}{(2x-1)(x^2+4)} = \frac{A}{2x-1} + \frac{Bx+C}{x^2+4}$$

$$x^2 - x - 21 = A(x^2+4) + (Bx+C)(2x-1)$$

$$x^2 - x - 21 = Ax^2 + 4A + 2Bx^2 - Bx + 2xC - C$$

Putting $x = \frac{1}{2}$ we get

$$-\frac{1}{4} - \frac{1}{2} - 21 = A\left(\frac{1}{4} + 4\right)$$

$$\boxed{A = -5}$$

Comparing coefficients of x^2

$$1 = A + 2B$$

$$\boxed{B = 3}$$

Comparing coefficients of x

$$-1 = -B + 2C$$

$$\boxed{C = 1}$$

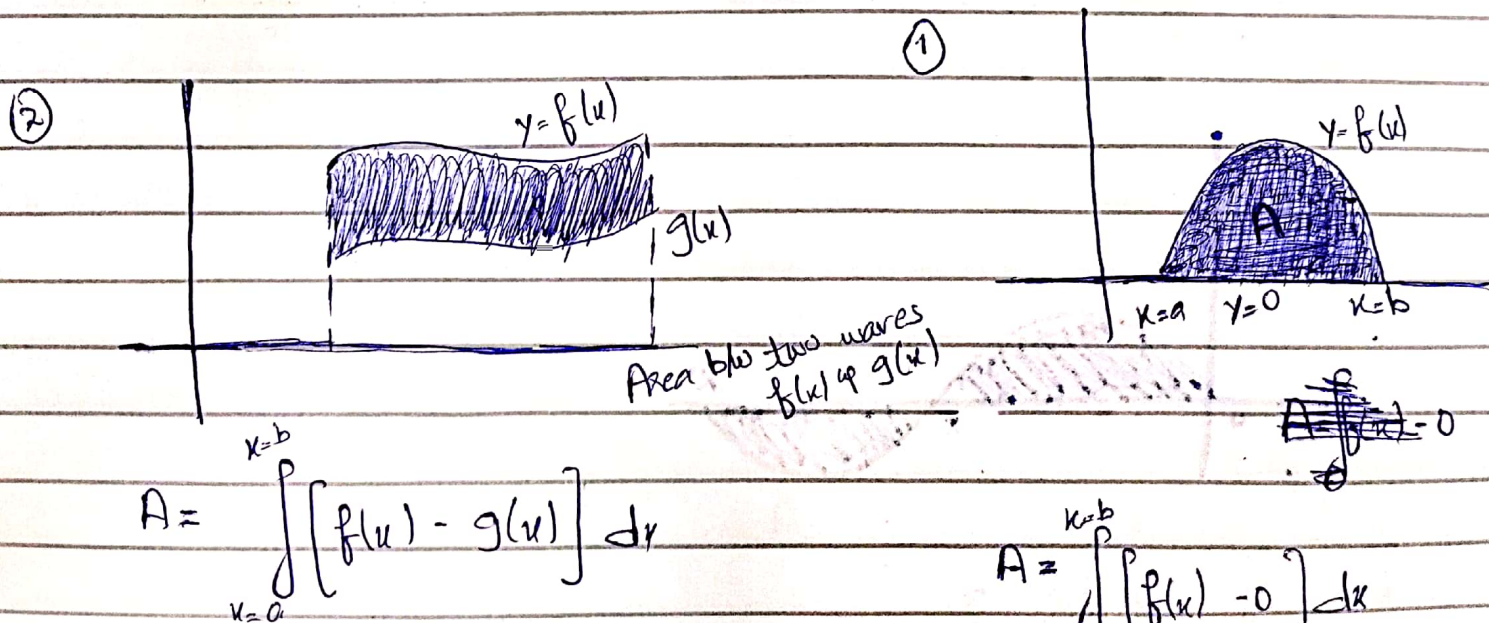
$$\int \left(\frac{A}{(2x-1)} + \frac{Bx+C}{x^2+4} \right) dx$$

$$= \int \left(\frac{-5}{2x-1} + \frac{3x+1}{x^2+4} \right) dx$$

$$= \frac{-5}{2} \int \frac{1 \times 2}{2x-1} dx + \frac{3}{2} \int \frac{2x}{x^2+4} dx + \int \frac{1}{x^2+4} dx$$

$$= \frac{-5}{2} \ln|2x-1| + \frac{3}{2} \ln|x^2+4| + \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C$$

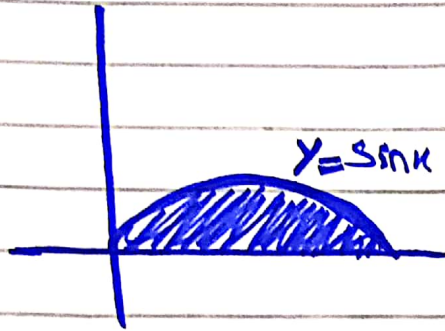
Area under the curve: or Area b/w the curve $y=f(x)$ and the x -axis.



$$A = \int_{x=a}^{x=b} [f(x) - 0] dx$$

$$A = \left| \int_{x=a}^{x=b} f(x) dx \right| \quad \text{or}$$

eg.



To find x-intercepts

$$\sin x = 0$$

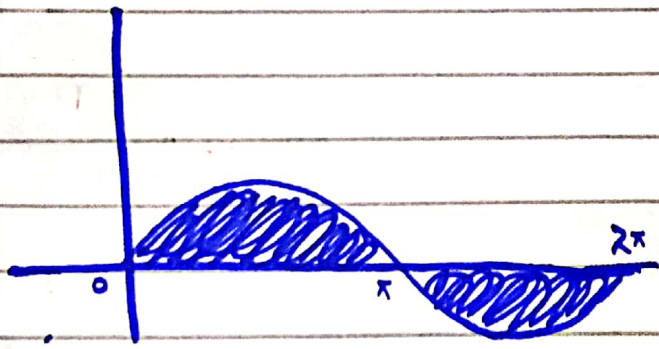
$$x = 0, \pi$$

$$A = \int_{x=0}^{x=\pi} \sin x \, dx$$

$$= -\cos x \Big|_0^{\pi}$$

$$= -[\cos \pi - \cos 0]$$

$$= 2$$



To find intercepts

$$\sin x = 0$$

$$x = 0, \pi, 2\pi$$

$$A_1 = \int_0^{\pi} \sin x \, dx = 2$$

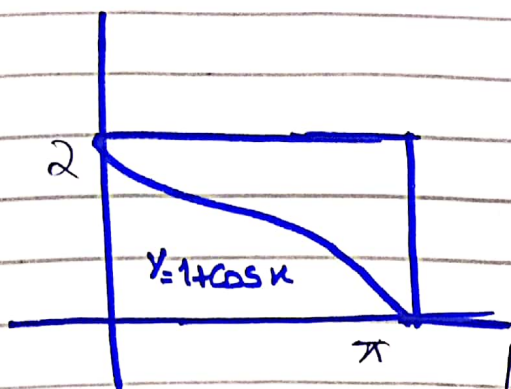
$$A_2 = \int_{\pi}^{2\pi} \sin x \, dx = -2$$

Total

$$= |A_1| + |A_2|$$

$$= |2| + |-2|$$

$$= 4$$



Area of Rectangle = 2π

Area b/w $y = 1 + \cos x$

and x -axis: $\int_0^{\pi} (1 + \cos x) dx$

$$= x + \sin x \Big|_0^{\pi}$$

$$= (\pi + \sin \pi) - (0 + \sin 0)$$

$$= \pi$$

Required Area

$$= |A_1| - |A_2|$$

$$= 2\pi - \pi$$

$$= \pi$$

OR

$$A = \int_0^{\pi} 2 - (1 + \cos x) dx$$

$$= \int_0^{\pi} (1 - \cos x) dx$$

$$= (x - \sin x) \Big|_0^{\pi}$$

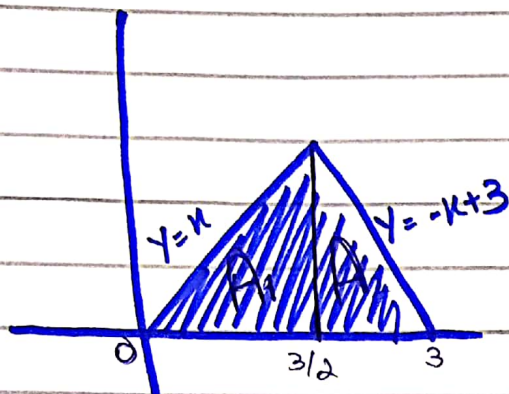
$$= (\pi - \sin \pi) - (0 - \sin 0)$$

$$= \pi$$

Ex 5-4
Q 63, 64 (lazmi tzy)
Ex 5-6
Q 42-64 and 65-66

→ lazmi
swal in paper

5 =



For x -intercepts

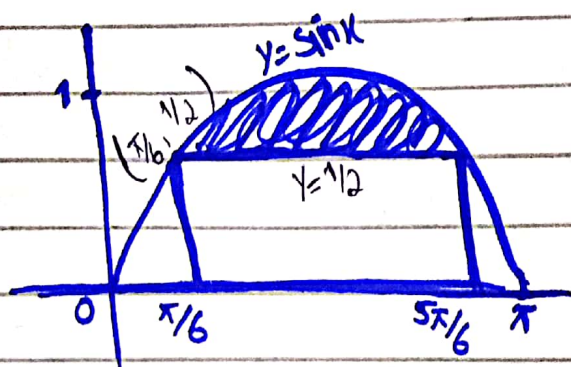
$$\begin{aligned} -x+3 &= 0 \\ x &= 3 \end{aligned}$$

$$\begin{aligned} x &= -x+3 \\ x &= 3/2 \end{aligned}$$

$$A_1 = \int_0^{3/2} x \, dx$$

$$A_2 = \int_{3/2}^3 (-x+3) \, dx$$

Total $A = |A_1| + |A_2|$



$$\int_{\pi/6}^{5\pi/6} (\sin x - 1/2) \, dx$$

$$5\pi/6 - \pi/6 = 2\pi/3$$

$$A_2 = \text{Area of Rectangle} = l \times w = 2\pi/3 \times 1/2 = \pi/3$$

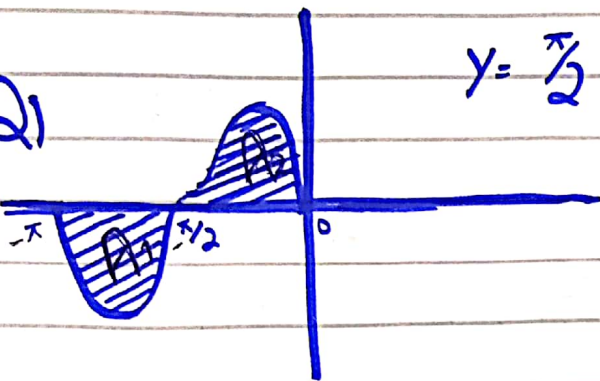
A_1 = Area under curve $y = \sin x$ and x -axis

$$A_1 = \int_{\pi/6}^{5\pi/6} (\sin x - 0) \, dx$$

$$A = |A_1| - |A_2| =$$

Ex 5.6 ⁴²⁻⁶⁴

Q1



$$y = \frac{\pi}{2} \cos k [\sin(\pi + \sin \pi k)]$$

$$\begin{aligned} \frac{\pi}{2} \cos k [\sin(\pi + \sin \pi k)] &= 0 \\ \cos k [\sin(\pi + \sin \pi k)] &= 0 \\ \cos k &= 0 \\ k &= -\pi/2 \\ k &= -3\pi/2 \end{aligned}$$

$$A_1 = \int_{-\pi}^{\pi/2} \left[0 - \frac{\pi}{2} \cos [\sin(\pi + \sin \pi k)] \right] dk$$

$$= -\frac{\pi}{2} \int_{-\pi}^{\pi/2} \cos \pi k \cdot \sin(\pi + \sin \pi k) dk$$

$$= + \frac{\pi}{2} \int_{-\pi}^{\pi/2} \cos k \cdot \sin(\sin \pi k) dk$$

$$\begin{aligned} &= \sin(\pi + \sin \pi k) \\ &= \sin \pi \cos(\sin \pi k) + \cos \pi \sin(\sin \pi k) \\ &= -\sin(\sin \pi k) \end{aligned}$$

$$= \frac{\pi}{2} \left[\sin(\sin \pi k) \cdot \sin k \right]_{-\pi}^{\pi/2} - \int_{-\pi}^{\pi/2} \sin k \cdot \cos(\sin \pi k) \pi \cos k dk$$

$$= \frac{\pi}{2} \left[\sin(\sin(\pi \cdot -\pi/2)) \cdot \sin(-\pi/2) - 0 - \pi \int_{-\pi}^{\pi/2} \sin k \cdot \cos \pi k \cdot \cos(\sin \pi k) dk \right]$$

$$= \frac{\pi}{2} \left[() - \pi \int_{-\pi}^{\pi/2} \sin k \cdot \cos \pi k \cdot \cos(\sin \pi k) dk \right]$$

where: $-\pi/2$

$$\int_{-\pi}^{\pi} \sin \kappa - \cos \pi \kappa \cdot \cos(\sin \pi \kappa) d\kappa$$



"

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إِنَّا لِلّٰهِ وَإِنَّا إِلَيْهِ رَاجِعُونَ

$$\kappa = -\pi/2$$

$$\int_{-\pi}^{\pi} \cos \pi \kappa \cdot \cos(\sin \pi \kappa) d\kappa$$

$$\kappa = -\pi$$

$$= \frac{1}{\pi} \int_{0.43}^{0.92} \cos u$$

$$= \frac{1}{\pi} \sin u \Big|_{0.43}^{0.92}$$

$$= \frac{1}{\pi} [\sin(0.92) - \sin(0.43)]$$

$$\text{let } u = \sin \pi \kappa$$

$$du = \pi \cos \pi \kappa d\kappa$$

$$u = \sin \pi \kappa$$

$$\text{where } \kappa = -\pi/2$$

$$v = \sin\left(\frac{-\pi^2}{2}\right)$$

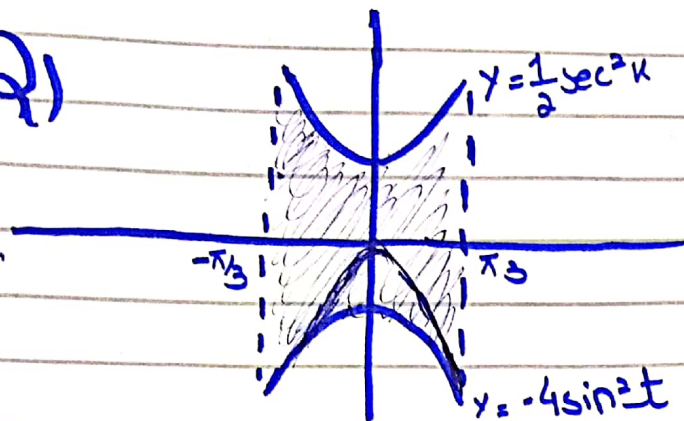
$$u = 0.92$$

$$\text{where } \kappa = -\pi$$

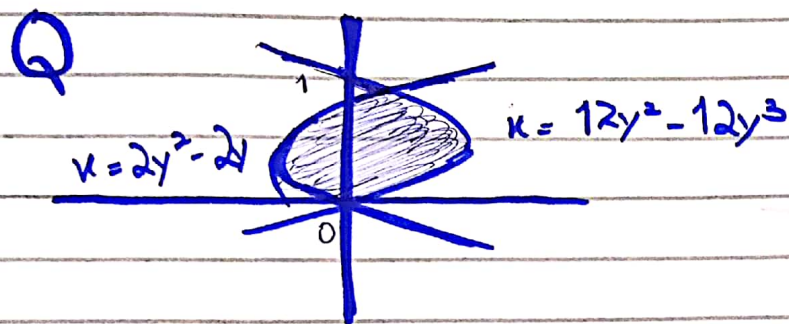
$$u = \sin(-\pi^2)$$

$$u = 0.43$$

Q1

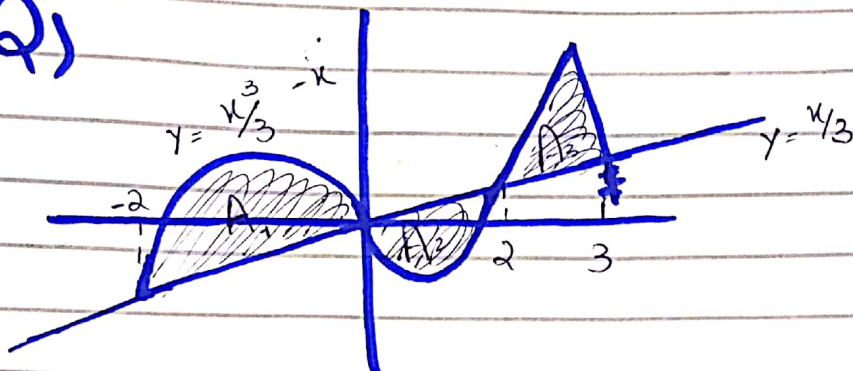


$$\begin{aligned}
 A &= \int_{-\pi/3}^{\pi/3} \left(\frac{1}{2} \sec^2 x + 4 \sin^2 x \right) dx \\
 &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} \sec^2 x dx + 4 \int_{-\pi/3}^{\pi/3} \sin^2 x dx \\
 &= \frac{1}{2} \tan x \Big|_{-\pi/3}^{\pi/3} + \frac{4}{2} \left[\int_{-\pi/3}^{\pi/3} (1 - \cos 2x) dx \right]
 \end{aligned}$$



$$A = \int_{y=a=0}^{y=b=1} [(12y^2 - 12y^3) - (2y^2 - 2y)] dy$$

Q1



$$A_1 = \int_{-2}^0 \left[\left(\frac{x^3}{3} - x \right) - \frac{x}{3} \right] dx$$

$$\frac{x^3}{3} - x = \frac{x}{3}$$

$$\frac{x^3}{3} - x - \frac{x}{3} = 0$$

$$x^3 - 3x - x = 0$$

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$x = 0, \quad x = -2, +2$$

$$A_2 = \int_0^2 \left[\left(\frac{x^3}{3} - x \right) - \frac{x}{3} \right] dx$$

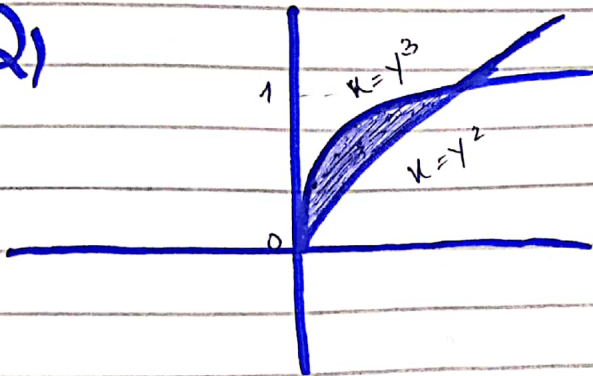
$$A_3 = \int_2^3 \left[\left(\frac{x^3}{3} - x \right) - \frac{x}{3} \right] dx$$

Total Area =

$$= |A_1| + |A_2| + |A_3|$$

=

Q1



$$A = \int_{y=0}^{y=1} (y^3 - y^2) dy$$

$$= \left(\frac{y^4}{4} - \frac{y^3}{3} \right) \Big|_0^1$$

$$= \frac{1}{4} - \frac{1}{3}$$

$$= \frac{3-4}{12} = -\frac{1}{12}$$

or in terms of x

$$\begin{aligned} x &= y^3 \\ y &= x^{1/3} \end{aligned}$$

$$\begin{aligned} x &= y^2 \\ y &= x^{1/2} \end{aligned}$$

$$A = \int f(x) - g(x)$$

$$= \int (x^{1/3} - x^{1/2}) dx$$

$$x^{1/3} - x^{1/2}$$

$$x^{1/3} - x^{1/2} = 0$$

$$x^{1/2}(-1 + x^{1/6}) = 0$$

$$x = 0, 1$$

• Improper Integrals * 1 question pka

Integral with infinite limits of integration are called improper integral.

① If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

② If $f(x)$ is continuous on $(-\infty, b]$ then

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

③ If $f(x)$ is continuous on $(-\infty, \infty)$

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx$$

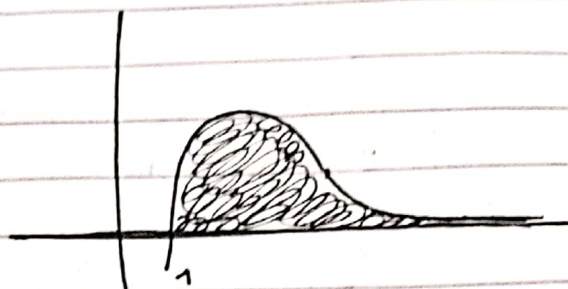
OR

$$= \lim_{a \rightarrow -\infty} \int_a^0 f(x) dx + \lim_{b \rightarrow \infty} \int_0^b f(x) dx$$

Ex $f(x) = \frac{\ln x}{x^2}$

$$A = \int_1^{\infty} \left(\frac{\ln x}{x^2} - 0 \right) dx$$

$$A = \lim_{b \rightarrow \infty} \int_1^b \frac{\ln x}{x^2} dx$$



where

$$I = \int_1^b \frac{\ln x}{x^2} dx$$

$$= \int_1^b x^{-2} \ln x dx$$

$$= \ln x \cdot \frac{x^{-1}}{-1} \Big|_1^b - \int_1^b \frac{x^{-1}}{-1} \cdot \frac{1}{x} dx$$

$$= -\frac{\ln x}{x} \Big|_1^b + \int_1^b x^{-2} dx$$

$$= -\frac{\ln x}{x} \Big|_1^b + \frac{x^{-1}}{-1} \Big|_1^b = -\left[\frac{\ln b}{b} - \frac{\ln(1)}{1} \right] - \left[\frac{1}{b} - \frac{1}{1} \right]$$

$$A = \lim_{b \rightarrow \infty} \int_1^b \frac{\ln x}{x^2} dx$$

$$= \lim_{b \rightarrow \infty} \left[-\left(\frac{\ln b}{b} - 0 \right) - \left(\frac{1}{b} - 1 \right) \right]$$

$$= -\lim_{b \rightarrow \infty} \frac{\ln b}{b} - \lim_{b \rightarrow \infty} \frac{1}{b} + 1$$

$$= 0 - 0 + 1$$

$$A = 1$$

$\Rightarrow$

$$\therefore \lim_{b \rightarrow \infty} \frac{\ln b}{b} \left(\frac{\infty}{\infty} \right)$$

Using L'Hopital's rule

$$= \lim_{b \rightarrow \infty} \frac{1/b}{1}$$

$$= \lim_{b \rightarrow \infty} \frac{1}{b} = 0$$

Notes

① An integral is called convergent if it has a finite value. ~~This means~~ or the area has finite value

② An integral is called divergent if its value is $+\infty$ or $-\infty$ or area has infinite value

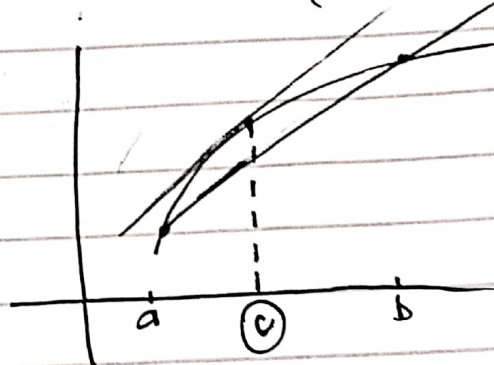
∴ 1 sural pka

Mean value theorem:

Suppose $y = f(x)$ is a continuous function over a close interval $[a, b]$ and differentiable on open interval (a, b) . Then there ~~is~~ ^{is} at least one point c in (a, b) such that or at which

$$\frac{f(b) - f(a)}{b - a} = f'(c) \text{ (average rate of change) = (instantaneous rate of change)}$$

∴ c is a point where slope of secant & slope of tangent will be equal



$$f(x) = x^2 + 2x - 1 \quad [0, 1]$$

$$= (x+1)^2 - 2$$

$$f'(x) = 2x + 2$$

$$f'(c) = 2c + 2$$

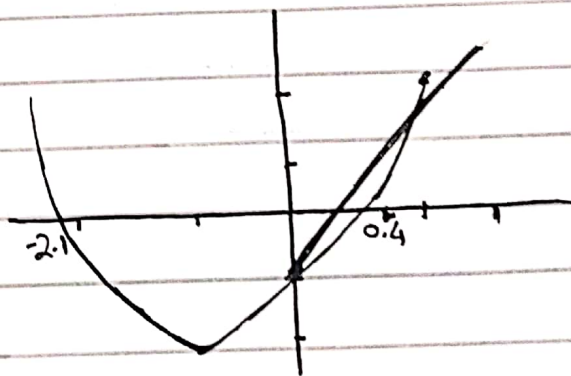
$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

$$\frac{f(1) - f(0)}{1 - 0} = 2c + 2$$

$$2 - (-1) = 2c + 2$$

$$2 + 1 = 2c + 2$$

$$\boxed{c = \frac{1}{2}}$$



$$Q) f(x) = \begin{cases} x^3 & -2 \leq x < 0 \\ x^2 & 0 \leq x \leq 2 \end{cases}$$

Case-I

$$f(x) = x^3 ; [-2, 0)$$

Case-II

$$f(x) = x^2 ; [0, 2]$$